# Characterization of studies with less than 10 patients that included the use of robotic-assisted surgery

| Author (Year) | Type of Study | Level of Evidence | Population | Cohort size | Control group | Mean age (% female) | Inclusion Criteria | Exclusion Criteria | Mean Follow-Up | Outcomes Measured | Results | Comments |
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| Chang et al., 2020 | Retrospective | IV | Sympathetic Nerve Reconstruction (SNR)<br><br>Chang Gung University, Taoyuan | 7 | N/A | Median: 41 years (29% F) | Post-operative compensatory sweating necessitating SNR | No history of non-surgical sympathectomy | F/u every 6 months postoperatively by phone | mortality, operating time, LOS, graft type, symptom improvement | Operative mortality (none)<br><br>Median operating time: 10.5h<br><br>LOS Median: 4 days<br><br>Symptom improvement of 70% | Short f/u period - could only assess symptom improvement in one patient |
| Asaad et al., 2021 | retrospective case series | IV | Patients who underwent robotic flap harvest following robotic extirpative surgery<br><br>Houston, TX | 7 | Open rectus abdominis muscle harvest (n=95) | 66 | Patients who underwent open or robotic flap harvest following robotic extirpative surgery | N/A | 16 ± 24 months (median, 8 months) | Outcomes/complications | No significant differences in complications and hernia recurrences between the robotic and open techniques | Small sample size, short f/u time, difference in surgical indications and defect types |
| Sieffert et al., 2017 | retrospective | | Patients with locally advanced low rectal cancer and underwent extralevator excision with gracilis flap reconstruction<br><br>Dayton, OH | 6 | N/A | 63 | Patients with locally advanced low rectal cancers not amenable to sphincter preservation | N/A | N/A | Demographic<br>Operative data<br>Outcomes/complications | No intraoperative perforation or positive circumferential resection margin.<br><br>Complications include locoregional recurrence, flap venous congestion, perineal abscess that went on to complete healing. | Small sample size, short f/u time |
| Haverland et al., 2021 | case series | IV | Robotic-assisted rectus abdominus harvest for pelvic floor reconstruction<br><br>Arizona, USA | 6 | N/A | 69.2 (66.6%) | Vesicovaginal fistula, pelvic organ prolapse, exenteration, vaginectomy, partial vulvectomy, and abdominoperineal resection. | Not reported | 9.2 mo. | Bowel obstructions, infections, emergency room visits, and readmission. | 1 patient: occasional abdominal pain<br><br>1 patient: intermittent bowel obstruction<br><br>1 patient: pelvic abscess, requiring readmission. | |
| Selber, 2010 | Case series | IV | Patients with SCC<br><br>Houston, TX | 5 | N/A | 70.8 | Patients with SCC | N/A | N/A | Outcomes/Complications | No intraoperative complications<br><br>Successful reconstruction | Small sample size precluded finding sig. difference in operative times/hospital stay |
| Singh et al., 2015 | case series | IV | Robotic extralevator abdominoperineal excision with robotic rectus abdominis flap harvest for distal rectal adenocarcinoma<br><br>Chicago, USA | 3 | N/A | 50.3 (33%) | Not reported | Not reported | Not reported<br><br>Longest: 16 mo. | Hernia or bulge formation<br><br>Infection | Early perineal wound infection<br><br>1 patient: parastomal hernia<br><br>1 patient: abdominal weakness | Average operative time: 522 min. |
| Vigneswaran et al., 2021 | Case Series | | Paraesophageal hernias<br><br>Chicago, USA | 2 | N/A | 70.5 (50%) | Paraesophageal hernia | N/A | 5 m | Technique,<br>Outcomes/complications | No signs of recurrence at 5 m f/u<br><br>No adverse effects of harvsting posterior rectus shealth | Short f/u; no abdominal CT scans for better post-op clinical evaluation and to evaluate role of mesh |
| Patel et al., 2012 | case report | IV | Rectus abdominis harvest for lower extremity defect<br><br>Ohio, USA | 1 | N/A | 30 (100%) | N/A | N/A | 6 mo. | Not reported | No hernia or bulge<br><br>No infections noted | Short followup and only 1 patient limits the validity of this study |
| Gundlapalli et al., 2018 | case report | IV | Robotic harvest of deep inferior epigastric perforator flap<br><br>South Carolina, USA | 1 | N/A | 51 (100%) | N/A | N/A | 9 mo. | Not reported | No flap or donor-site complications<br><br>No hernia or bulge | Setup time: 20 min.<br><br>Robotic pedicle dissection: 40 min.<br><br>Traditional DIEP cost: $14,800<br><br>Robotic DIEP cost: $16,300 |
| Özkan et al., 2019 | case report | IV | Robotic harvest of omental flap for extremity coverage<br><br>Istanbul, Turkey | 1 | N/A | 58 (0%) | N/A | N/A | 12 mo. | Not reported | No flap or donor-site complications<br><br>No infection | Total operative time: 2.5 hrs |
| Day et al., 2021 | case report | IV | Robotic harvest of a pedicled omentum flap for chest wall defect<br><br>Minnesota, USA | 1 | N/A | 68 (100%) | N/A | N/A | 2 mo. | Disease recurrence<br><br>Flap or donor-site complications | No intra- or postoperative complications<br><br>Patient died 3 months postop due to metastatic disease | |

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| Facca & Liverneaux, 2010 | Case Report | IV | Man with hypothenar hammar syndrome<br><br>Strasbourg, France | 1 | N/A | 39 | Hypothenar hammar syndrome | N/A | 6 months | Postoperative after-effects<br><br>Improvement in paresthesias, vasomotor disorders, and cold-related pain episodes | No postoperative after-effects reported<br><br>Vasomotor disorders disappeared immediately after surgery<br><br>Progressive regression of paresthesis in L ulnar nerve after 2m<br><br>Some infrequent persistant cold-related pain episodes at 6m postoperatively | |
| Naito et al., 2020 | Case Report | IV | Patient with brachial plexus injury<br><br>Tokyo, Japan | 1 | N/A | 57 | Patient with brachial plexus injury, need for intercostal n. harvest | N/A | N/A | Outcomes/complications | No respiratory complications, no pain at nerve-harvesting site; patient could ambulate freely POD 1; good patient satisfaction | Specialized technique, small sample size |
| Yilmaz et al., 2020 | Case Report | IV | Patient with need for penile replantation<br><br>Ankara, Turkey | 1 | N/A | 35 | Patient with need for penile replantation | N/A | 6 weeks | Outcomes/complications | Patient could form an erection, no stricture or fiscula on voiding cystourethryrogram<br><br>No hematoma or glans necrosis, some necrotic penile skin was debrided and recon. with a fasciocutaneous flap | Small sample size |
| Teven et al., 2021 | Case Report | IV | Patient with lower extremity lymphedema<br><br>Pheonix, Arizona | 1 | N/A | 52 | Patient with need for omentum lymphatic transplant | N/A | 10 weeks | Outcomes/complications, operative time | Robotic harvest: 80 mins<br><br>100mL blood loss<br><br>Patient discharged 24 h after surgery with full ambulation<br><br>At 10 wk, incisions healed, sig. improvement in lymphedema symptoms | Small sample size, short f/u time |
| Chang et al., 2021 | Case Report | IV | Patient with palmar hyperhidrosis<br><br>Taoyuan, Taiwan | 1 | N/A | 59 | Patient with need for sympathetic trunk reversal reconstruction with sural n. graft | N/A | 42 months | Outcomes/complications | At 24, 33, and 42 m postoperatively, symptoms improved by 70%<br><br>No intra or post-operative complications, pt discharged POD 4 | Robot use requires healthy n. stump, reliable technique in order to be safe |